

CHAPTER FIVE

COMMON BUSINESS APPLICATIONS OF

INFORMATION TECHNOLOGY

Chapter Outlines

5.1 Financial Information Systems

5.2 Marketing Information Systems

5.3 Manufacturing and Production Information
Systems

5.4 Human Resource Information Systems

5.5 Managerial Decision Support Systems

5.6 Transaction Processing System (TPS)

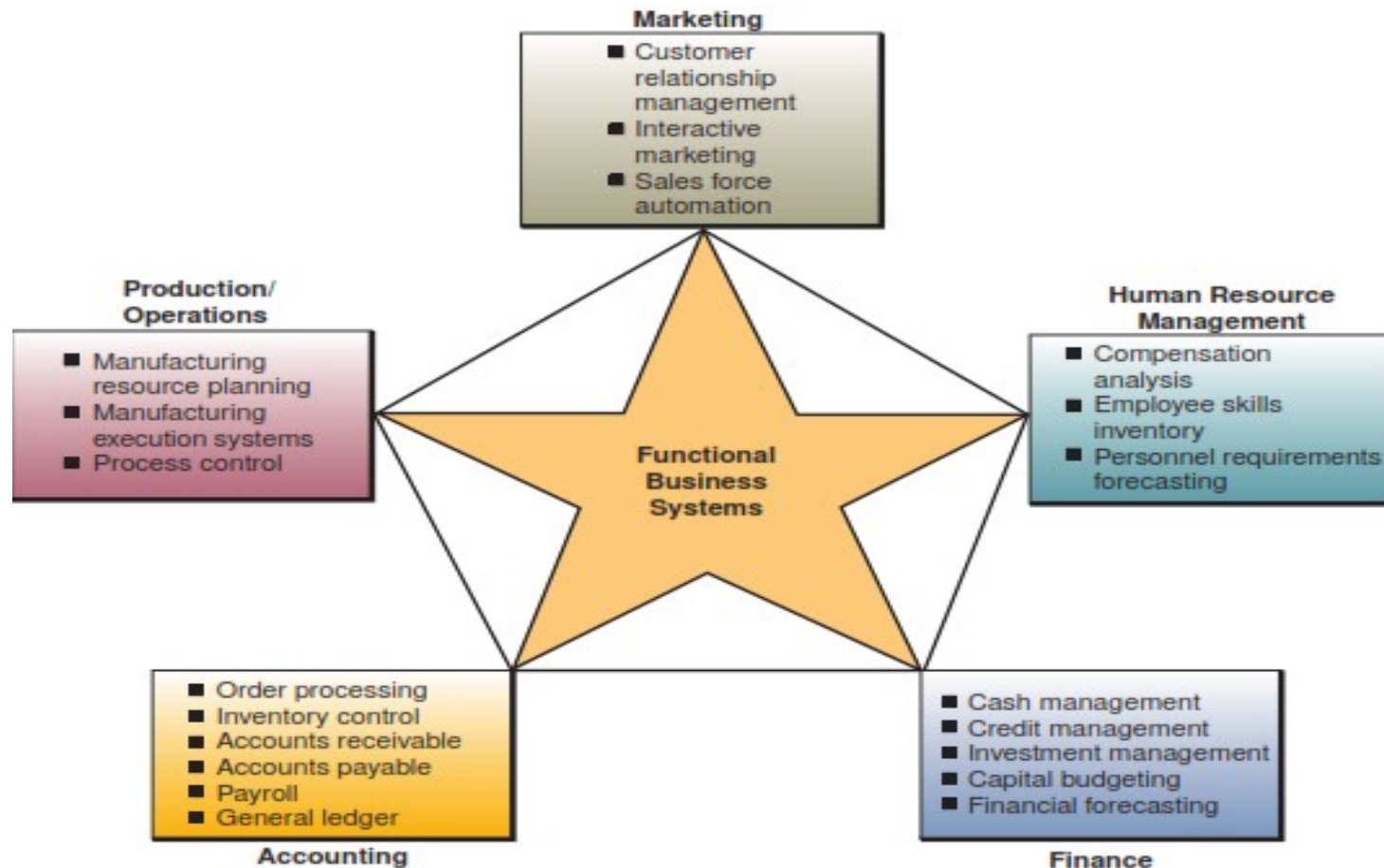
5.7 E-Business

- *Business managers are moving from a tradition where they could avoid, delegate, or ignore decisions about IT to one where they cannot create a marketing, product, international, organization, or financial plan that does not involve such decisions.*
- There are as many ways to use information technology in business as there are
 - business activities to be performed,
 - business problems to be solved, and
 - business opportunities to be pursued.
- As a business professional, you should have a basic understanding and appreciation of the major ways information systems are used to support each of the functions of business that must be accomplished in any company that wants to succeed.

- As a business professional, it is also important that you have a specific understanding of how information systems affect a particular business function (e.g., marketing) or a particular industry (e.g., banking) that is directly related to your professions.
- For example, someone whose job is a marketing position in banking should have a basic understanding of how information systems are used in banking and how they support the marketing activities of banks and other firms.

The following figure illustrates how information systems can be grouped into business function categories.

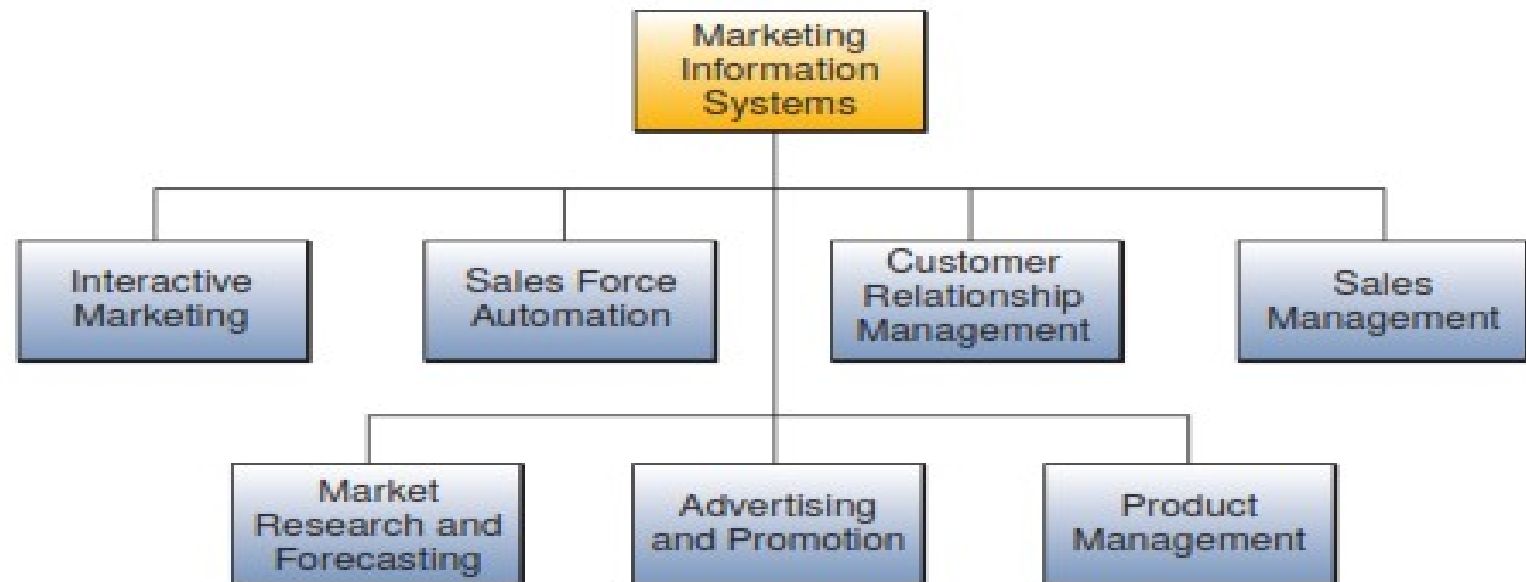
Figure 5.1 Examples of functional business information systems.



5.2 Marketing Information Systems

- The business function of marketing is concerned with the planning, promotion, and sale of existing products in existing markets, and the development of new products and new markets to better attract and serve present and potential customers.
- Marketing information systems provide information technologies that support major components of the marketing function.
- Figure 5.2 illustrates how marketing information systems provide information technologies that support major components of the marketing function.

- For example, Internet/intranet web sites and services make an *interactive marketing* process possible where customers can become partners in creating, marketing, purchasing, and improving products and services.
- *Sales force automation* systems use mobile computing and Internet technologies to automate (computerize) many information processing activities for sales support and management.



6.3 Manufacturing and Production Information System

- Manufacturing information systems support the *production/operations function that includes* all activities concerned with the planning and control of the processes producing goods or services.
- A variety of manufacturing information systems, many of them Web-enabled, are used to support ***computer-integrated manufacturing (CIM)***.

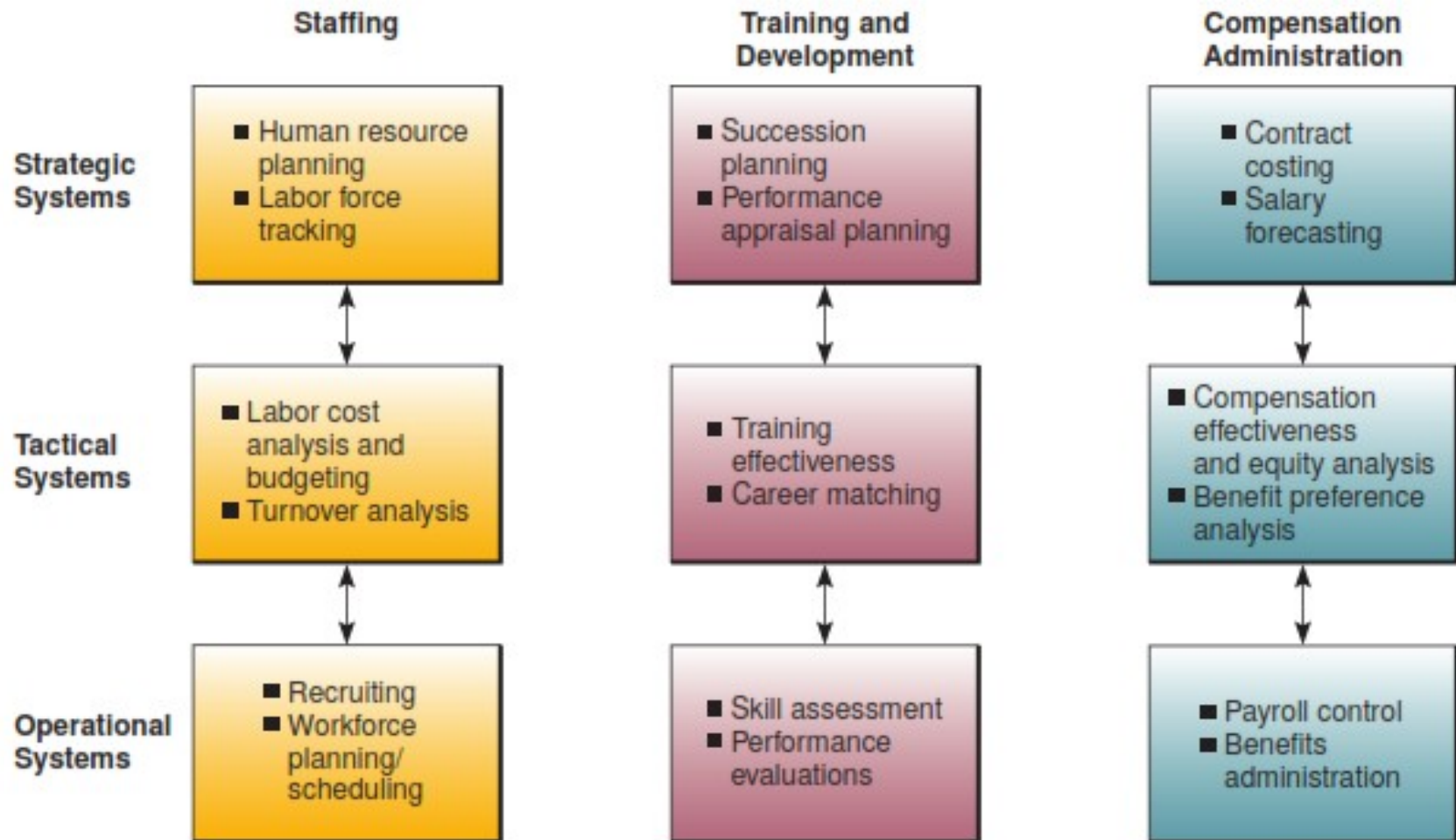
- CIM is an overall concept that emphasizes that the objectives of computer-based systems in manufacturing must be to:
 - ***Simplify (reengineer)***: production processes, product designs, and factory organization as a vital foundation to automation and integration.
 - ***Automate***: production processes and the business functions that support them with computers, machines, and robots.
 - ***Integrate***: all production and support processes using computer networks, cross-functional business software, and other information technologies.
- The overall goal of CIM and such manufacturing information systems is to produce products of the highest quality.

6.4 Human Resource Information System

- The human resource management (HRM) function involves the recruitment, placement, evaluation, compensation, and development of the employees of an organization.
- The goal of human resource management is the effective and efficient use of the human resources of a company.
- The Internet has become a major force for change in human resource management.
- *For example,* online HRM systems may involve recruiting for employees through recruitment sections of corporate Web sites.

- Companies are also using commercial recruiting services and databases on the World Wide Web, posting messages in selected Internet newsgroups, and communicating with job applicants via e-mail.
- The Internet has a wealth of information and contacts for both employers and job hunters.
- Top Web sites for job hunters and employers on the World Wide Web include Monster.com, HotJobs.com and CareerBuilder.com

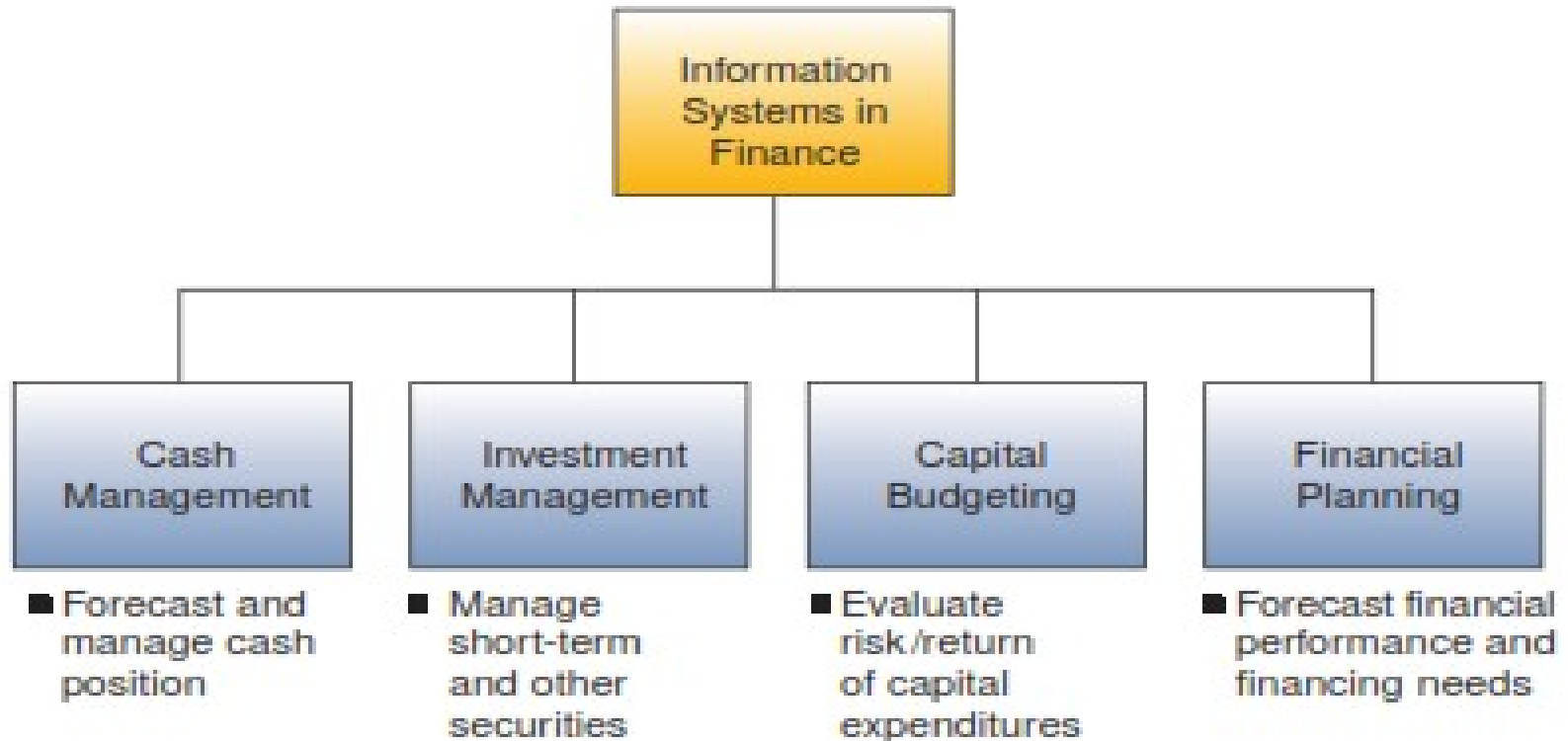
Figure 5.3 Human resource information systems support the strategic, tactical, and operational use of the human resources of an organization.



6.5 Financial Information System

- Computer-based financial management systems support business managers and professionals in decisions concerning
 - (1)the financing of a business and
 - (2)the allocation and control of financial resources within a business.
- Major financial management system categories include cash and investment management, capital budgeting, financial forecasting, and financial planning.
- Electronic spreadsheet packages, DSS software, and Web-based Groupware can be used for Financial Information System.

Figure 5.4 Examples of important financial management systems



5.6 Decision Support System (DSS)

- The success of an organization largely depends on the quality of the decisions that its employees make.
- When decision making involves large amounts of information and a lot of processing, computer-based systems can make the process efficient and effective.
- When do you have to make a decision? When you drive your car to a certain destination and there is only one road, you do not have to make a decision. The road will take you there.
- But if you come to a fork, you have to decide which way to go. In fact, whenever more than one possible action is available, a decision must be made.

- If you have to decide based only on distance, making a decision is easy. If you have to choose between a short but heavily trafficked road and a longer road with lighter traffic, the decision is a bit more difficult.
- A decision is easy to make when one option will clearly bring about a better outcome than any other. Decisions become more difficult when more than one alternative seems reasonable and when the number of alternatives is great.
- In business, there can be dozens, hundreds, or even millions of different courses of action available to achieve a desired result. The problem is deciding on the best alternative.

- Decision Support Systems (DSS) are a class of computerized information systems that support decision-making activities.
- DSS are interactive computer-based systems and subsystems intended to help decision makers use communications technologies, data, documents, knowledge and/or models to successfully complete decision process tasks.
- Decision Support Systems should be defined as a broad category of information systems for informing and supporting decision-makers.
- DSS are intended to improve and speed-up the processes by which people make and communicate decisions.

Basic themes of DSS

- Information systems.
- Used by managers.
- Used in making decisions.
- Used to support, not to replace people.
- Used when the decision is
- "semi-structured" or "unstructured."
- Incorporate a database of some sort.
- Incorporate models.

6.7 Transaction Processing System (TPS)

- Whenever two people make an exchange, it is called a transaction.
- Transactions are important events for a company, and collecting data about them is called transaction processing.
- Examples of transactions include making a purchase at a store, withdrawing money from a checking account, making a payment to creditor, or paying an employee.
- Because transactions generally involve an exchange of money, it is critical that the data be protected during transmission and stored carefully so that it cannot be altered.
- It is also critical that the data be saved so that managers can verify the data if any conflicts arise.

- A transaction processing system (TPS) is an information system that records company transactions (a transaction is defined as an exchange between two or more business entities).
- Transaction processing systems (TPS) are cross-functional information systems that process data resulting from the occurrence of business transactions.
- Transactions are events that occur as part of doing business, such as sales, purchases, deposits, withdrawals, refunds, and payments.

- Let us look at a simple *example of a business transaction*.
- *McDonald's*, which sells a large number of hamburgers every day, orders raw materials from its suppliers. Each time the company places an order with a supplier, a transaction occurs and a transaction system records relevant information, such as the supplier's name, address, and credit rating, the kind and quantity of items purchased, and the invoice amount.

Process of Transaction Processing System

The five steps in processing a transaction are:

1. Data entry
2. Transaction Processing
3. Database Maintenance
4. Document and Report Generation
5. Inquiry Processing

1. Data Entry

- The first step of the transaction processing cycle is the capture of business data.
- There are a number of input devices *for entering data*:
 - Keyboard/video display terminals
 - Optical character recognition (OCR) devices, such as optical scanning wands and grocery check--*out scanners*.
 - Other technologies, including electronic mice, light pens, magnetic stripe cards, and voice input.

2. Transaction Processing

- Transaction processing systems process data in two basic ways:
 - (1) batch processing, where transaction data are accumulated over a period of time and processed periodically, and
 - (2) real-time processing (also called online processing), where data are processed immediately after a transaction occurs.
- All online transaction processing systems incorporate real-time processing capabilities.

3. Database Maintenance

- An organization's databases must be updated by its transaction processing systems so that they are always correct and up-to-date.
- For example, credit sales made to customers will cause customer account balances to be increased and the amount of inventory on hand to be decreased.
- Database maintenance ensures that these and other changes are reflected in the data records stored in the company's databases.

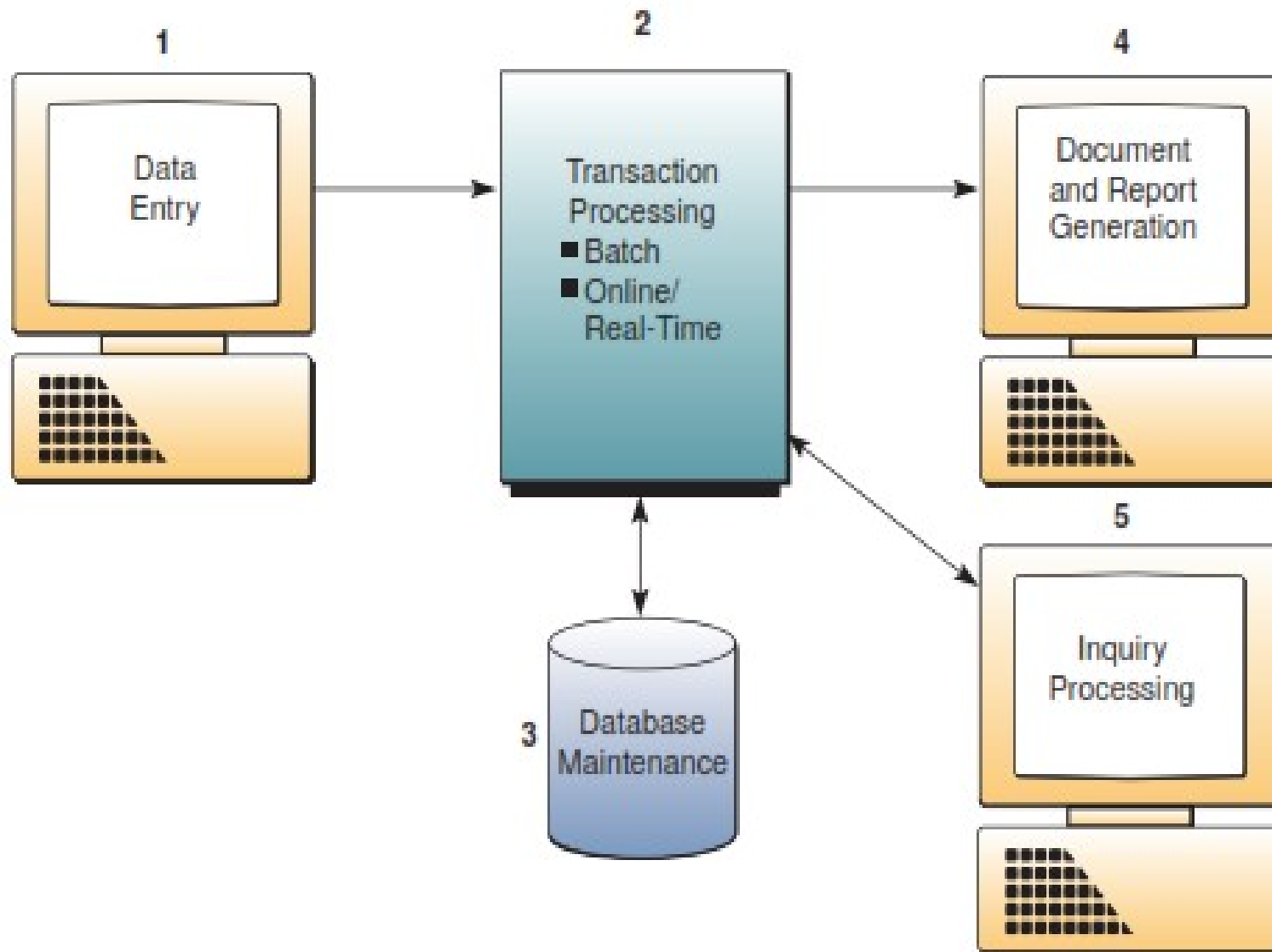
4. Document and Report Generation

- Transaction processing systems produce a variety of documents and reports.
- Examples of transaction documents include purchase orders, paychecks, sales receipts, invoices, and customer statements.
- Transaction reports might take the form of a transaction listing such as a payroll register.

5. Inquiry Processing

- The last step in processing a transaction is querying (asking questions of) the system.
- Query facilities allow users to process data and information that may otherwise not be readily available.
- For example, a sales manager may query the system the number of damaged items in a given store.

- Many transaction processing systems responses are displayed in a variety of pre-specified formats or screens.
- For example, you might check on the status of a sales order, the balance in an account, or the amount of stock in inventory and receive immediate responses at your PC.



6.7 E-Business (Electronic Business)

E-commerce is changing the shape of competition, the speed of action, and the streamlining of interactions, products, and payments from customers to companies and from companies to suppliers.

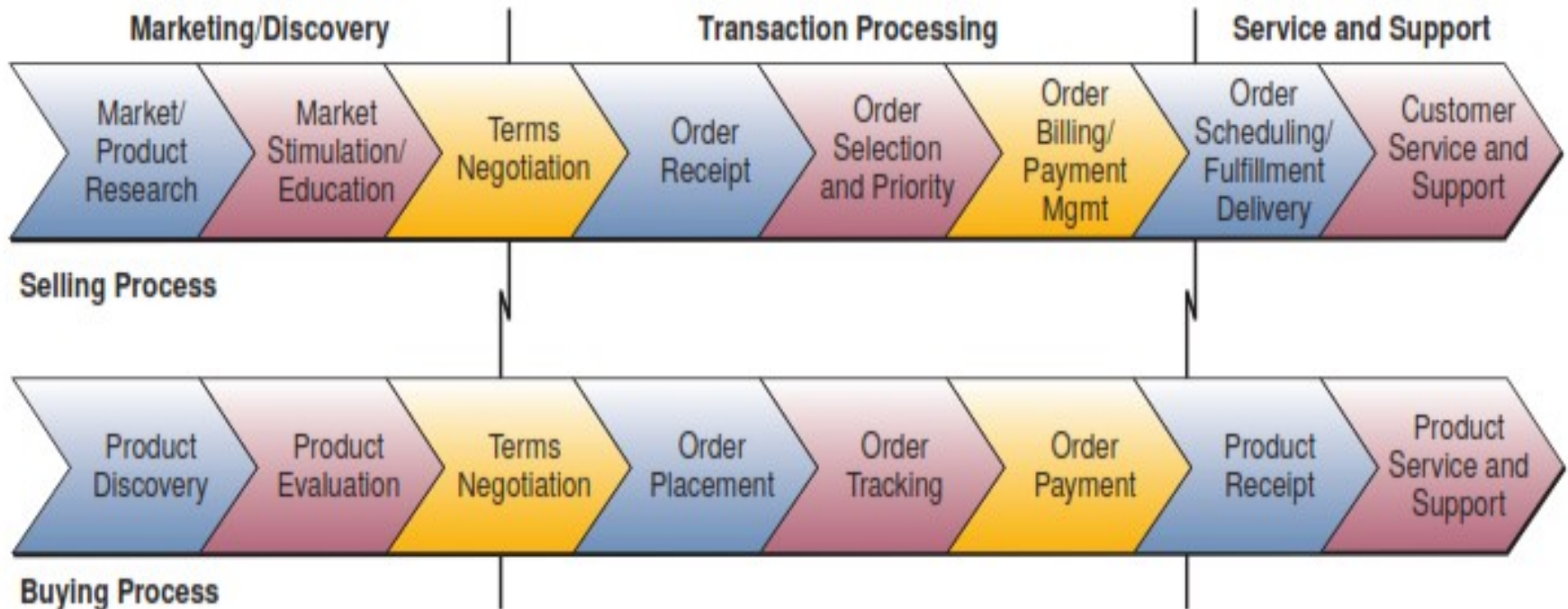
- **Electronic business, or e-business**, refers to the use of digital technology and the Internet to execute the major business processes in the enterprise.
- **E-business** includes activities for the internal management of the firm and or coordination with suppliers and other business partners. It also includes electronic commerce, or e-commerce.
- **E-commerce** is the part of e-business that deals with the buying and selling of goods and services over the Internet.
- It also encompasses activities supporting those market transactions, such as advertising, marketing, customer support, security, delivery, and payment.

- E-commerce refers to aspects of online business involving exchanges among customers, business partners and vendors.
- For example, suppliers interact with manufacturers, customers interact with sales representatives and shipment providers interact with distributors.
- E-business encompasses these elements, but also includes operations that are handled within the business itself.
- For example, production, development, corporate infrastructure and product management are aspects of e-business not included under the category of e-commerce.

- E-business and e-commerce have increased the *speed* and *ease* with which business can be transacted, resulting in intense competition among online vendors.
- To remain viable, e-businesses must adjust to *evolving technologies, continually integrate new systems* and *satisfy a wide variety of consumers*.
- If a business fails to do so, its customers do not have far to go to buy from competitors.

- The construction and maintenance of an e-business, especially one that processes a large number of transactions, requires
 - technical,
 - marketing and
 - advertising expertise.
- Customers want access to products and services on 24/7 basis (24 hours per day, 7 days per week).
- They also expect reliable, functional, fast and user-friendly services; companies that provide such services have higher success rates.

- Figure 5.6 E-commerce involves accomplishing a range of business processes to support the electronic buying and selling of goods and services.



Types of e-commerce

- E-commerce can be broken into four main categories: B2B, B2C, C2B, and C2C.

1. B2B (Business-to-Business)

- Companies doing business with each other such as manufacturers selling to distributors and wholesalers selling to retailers.
- Pricing is based on quantity of order and is often negotiable.

2. B2C (Business-to-Consumer)

- Businesses selling to the general public typically through catalogs utilizing shopping cart software. By dollar volume, B2B takes the prize, however B2C is really what the average people has in mind with regards to e-commerce as a whole.

3. C2B (Consumer-to-Business)

- A consumer posts his project with a set budget online and within hours companies review the consumer's requirements and bid on the project.
- The consumer reviews the bids and selects the company that will complete the project.

4. C2C (Consumer-to-Consumer)

- There are many sites offering free classifieds, auctions, and forums where individuals can buy and sell thanks to online payment systems like PayPal where people can send and receive money online with ease.
- eBay's auction service is a great example of where person-to-person transactions take place everyday since 1995.

E – COMMERCE ADVANTAGES:

- ❖ Faster buying/selling procedure, as well as easy to find products
- ❖ Buying/selling 24/7.
- ❖ More reach to customers, there is no theoretical geographic limitations.
- ❖ Low operational costs and better quality of services.
- ❖ No need of physical company set-ups.
- ❖ Easy to start and manage a business.
- ❖ Customers can easily select products from different providers without moving around physically.

E – COMMERCE DIS-ADVANTAGES:

- ❖ Any one, good or bad, can easily start a business. And there are many bad sites which eat up customers' money.
- ❖ There is no guarantee of product quality.
- ❖ Mechanical failures can cause unpredictable effects on the total processes.
- ❖ As there is minimum chance of direct customer to company interactions, customer loyalty is always on a check.
- ❖ There are many hackers who look for opportunities, and thus an e-commerce site, service, payment gateways, all are always prone to attack.

Build an e-commerce site

- Register a Domain name
 - Search for domain names
 - What is the criteria of selecting a good name?
- Find a Web Host
 - Select the suitable ISP
- Web Developers
 - Hire a web designer
 - Cut cost and work with the designer
 - If your site is easy to use, the customers will stay longer and come more frequent.

Online shopping risks

- A list of safeguards to keep in mind.
 - Shop with trusted merchants.
 - Read the site's delivery, return, and privacy policies.
 - Only enter your details with the URL that begins with [https://](#)
 - Be careful not to hit the ORDER NOW button more than once.
 - Never send credit card information via e-mail.
 - Keep a record of your transactions until shipped.